Storage Classes and Persistent Volume PV are Cluster Resources
Persistent Volume Claim PVC and Application are Project/Namespace Resources

Cluster Administrator: Storage Class: Not Namespace/Project
Cluster Administrator: Persistent Volume: Not Namespace/Project
Developer: Persistent Volume Claim: Namespace/Project
Developer: Application : Namespace/Project

RWX
RWO
RM

Storage Classes
NFS (FileSystem[DEFAULT] XFS EXT4), GlusterFS, Ceph, Azure,
Block(raw Disk) Storage, Local Volume LVM, iSCSI, AWS, Fibre Channel

Cluster Administrator

PV is not specific to Namespace/Project

Persistent Volume
PV
10Gi
BOUND

Persistent Volume
PV
25Gi
Available

Persistent Volume
PV
15Gi
Available

Cluster Administrator
Not Namespace/Project

When PVC requires a PV,
1. Static/Manual create PV
2. Dynamic Creation PV
(Storage Classes) Automatically

Persistent Volume Claim
PVC
9Gi

Persistent Volume
PV
30Gi

Persistent Volume Claim
PVC
30Gi
Pending

Application Request/
Developer
Namespace/Project

PVC is specific to Namespace/Project

Application
Storage PVC
PersistentVolumeClaim

Developer

PV once used/bounded by one PVC , can not be used/bounded by another PVC